

Scientific Theories

BUILD VOCABULARY

ACADEMIC WORDS A

scientific **theory** describes a well-tested explanation for a range of phenomena. Scientific theories are different from scientific laws, and it is important to understand that theories do not become laws. Laws, such as ideal gas laws in chemistry or Newton's laws of motion, are concise, specific descriptions of how some aspects of the natural world are expected to behave in a certain situation. In contrast, scientific theories, such as cell theory or the theory of evolution, are more dynamic and complex. Scientific theories encompass a greater number of ideas and hypotheses than laws, and are constantly fine-tuned through the process of science.

Researchers hope that data from many individual experiments add up to something bigger—a larger and more useful understanding of how the world works. That's where we encounter an important scientific term: **theory**. Many other terms you'll learn this year will be new to you, because they are used only in science. But the word *theory* is also often used in everyday life. That causes problems, because non-scientists use the word *theory* in a very different way than scientists do. When most people say, "I have a theory," they mean, "I have a hunch" or "I have a guess." When a friend says, "That's just a theory," she may mean, "People aren't too certain about that idea." Scientists would never use the word *theory* in that way. In those situations, a scientist would use the common words "guess" or "hunch" or a scientific term we've already discussed: *hypothesis*.

When scientists refer to gravitational theory or evolutionary theory, they do not mean "a hunch about gravity" or "a guess about evolution." **Q In science, the word theory applies to a tested, highly-reliable scientific explanation of events in the natural world that unifies many repeated observations and incorporates durable, well-supported hypotheses that enable scientists to make accurate predictions.** Charles Darwin developed lots of hypotheses over many years. It took a long time for him to assemble his thoughts and hypotheses into his theory of evolution by natural selection. Since then, evolutionary theory has predicted things Darwin couldn't have imagined, such as the evolution of bacteria that resist antibiotics and insects that are immune to pesticides. Today, evolutionary theory is the central organizing principle of all biological science.

Once a theory has been thoroughly tested and supported by many lines of evidence, it may become the dominant scientific view. But remember that no theory is absolute truth. Science is always changing; as new evidence is uncovered, a theory may be revised or replaced by a more useful explanation.



LESSON 1.1 Review

Q KEY IDEAS

1. In your own words, define the term *science*.
2. Why are hypotheses so important to controlled experiments?
3. How does a theory differ from a hypothesis?

CRITICAL THINKING

4. **Form a Hypothesis** You observe mold growing on one side of a slice of bread, but not on the other side. Form a hypothesis to explain this difference in mold growth.

5. **Plan an Investigation** Design a controlled experiment to test the effect of water temperature on goldfish. Be sure to include your hypothesis, independent variable, and dependent variable, as well as the experimental and control groups in your experiment.